

अजय के. सूद

भारत सरकार के प्रमुख वैज्ञानिक सलाहकार

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Dated: 29th April 2026

Dear *Dr. Mohanty,*

I am writing to you with reference to the letter (**enclosed**), which has also been addressed to you, from Shodh AI, a start-up selected under the IndiaAI mission, regarding the Large Physics Model (LPM) engineered by them.

As mentioned in the letter, the AI model is said to be able to generate exact, optimised 3D manufacturing blueprint for reactor components for advanced Stage-2/Stage-3 nuclear reactor technologies which are based on liquid sodium/molten salt coolants. Further, it is stated by the model developers that through the "Inverse Design" feature, the model will be able to generate 3D manufacturing blueprints of reactor components with mathematical guarantee for meeting the physical operational constraints, thus avoiding delays faced with the use of traditional simulator tools used for component design.

The proposal highlights a potential for accelerating India's 3-stage nuclear program by leveraging capabilities unlocked by advancements in Artificial Intelligence, to leapfrog our way to thorium utilisation for ensuring long-term energy security for the nation. We have studied the proposal inhouse and the methodology merits consideration for a deeper investigation by experts in your department.

Warm regards,

Yours sincerely,

(Ajay K. Sood)

Dr. Ajit Kumar Mohanty

Secretary, DAE

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Copy to;

CEO, Shodh AI